

Poetry Space: Visualization of Poetry Recitation Emotions in VR

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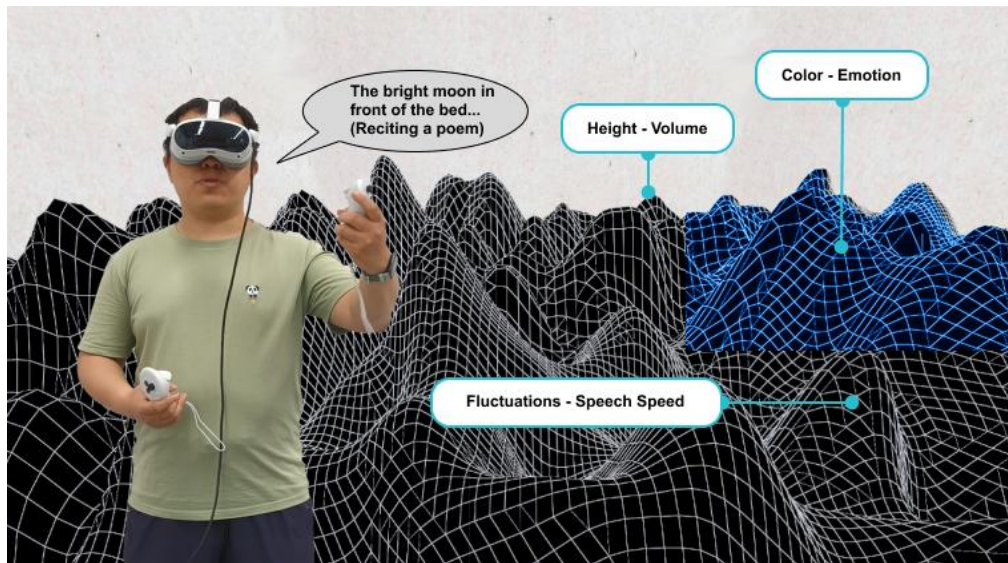


Figure 1: The VR experience of users in Poetry Space.

Abstract

Traditional pedagogical formats struggle to provide an audiovisual environment dynamically linked to recitation behavior to facilitate learner immersion. This paper explores emotion visualization for poetry recitation training in virtual reality (VR). We designed Poetry Space, which integrates a speech emotion recognition model to capture the reciter's emotions, volume and rhythm in real-time, dynamically transforming them into a virtual landscape of abstract

mountains. This provides objective, real-time and immersive emotional feedback for recitation pedagogy and training.

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1 INTRODUCTION

Recitation serves as a vital method for poetry learning, where the core of teaching and appreciation lies in the resonance of emotion and artistic conception. Classical Chinese poetics features the theory of "seeking spiritual essence through phonetic elements such as cadence, pace, and pitch". However, current recitation training faces limitations including subjective evaluation of emotional expression, lack of real-time feedback, and insufficient immersion in

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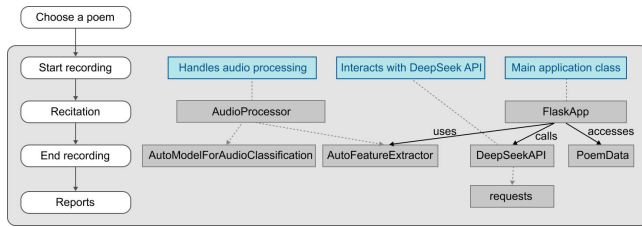


Figure 2: Technical workflow

artistic conception. Although "VR + language learning" research has demonstrated potential for contextual creation [Liu et al. 2024], it fails to address the intelligent needs of poetry recitation, which relies heavily on emotional and aesthetic experiences. With advances in large language models (LLMs) and affective computing, emerging studies explore AI-driven emotional support [Erdemir et al. 2023] or interactive narratives [Liang et al. 2024], presenting new opportunities. Thus, we designed Poetry Space (Figure 1), aiming to propose an innovative training method for poetry recitation through affective computing and multimodal interaction. Our contributions are:

- Speech emotion recognition and visualization: We synchronously transform multidimensional emotional features from speech signals into dynamic 3D landscape, creating an immersive recitation training space.
- Theoretical integration: We merge the classical Chinese poetics concept "seeking spiritual essence through phonetic elements" with embodied cognition theory, constructing a bidirectional feedback loop between recitation emotions and virtual landscapes in VR.

2 DESIGN AND IMPLEMENTATION

Poetry Space is deployed on the Pico 4 Ultra platform. Users select a poem from the library and begin reciting. Upon initiating recording, the system generates a mountain landscape synchronized with their voice, enabling users to adjust their delivery techniques based on dynamic visual feedback. After recitation, the system compares the emotional data with AI emotion model and generates an analytical report. The system supports both Chinese and English.

2.1 Emotion Recognition

We utilize the speech-emotion-recognition-with-openai-whisper-large-v3 model. Fine-tuned to identify vocal emotions, it aligns with AI-identified affective features of poems. The system workflow is shown in Figure 2.

2.2 Visualization Design

Poetry recitation demands exquisite control of emotion, breath, and rhythm. Traditional waveforms display physical sound parameters but lack aesthetic and cultural significance. We selected "mountains" as the core visual metaphor: in Chinese tradition, landscapes embody scholars' spirit, vitality, and emotions, while undulating mountain forms inherently mirror the dynamics of vocal rhythm. A triple-mapping system translates recitation emotions into an immersive 3D space:

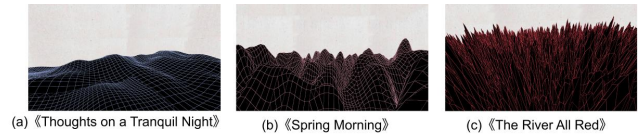


Figure 3: Participants recite poems of differing emotional tones

- Height $\Leftrightarrow$ Volume: Vocal intensity determines height (loudness - steep peaks; softness - gentle valleys).
- Color $\Leftrightarrow$ Emotion: Primary emotion categories dictate the scene's color palette, inspired by Chinese aesthetics. (Cool tones - sorrow, warm tones - excitement, consistent with temperature-emotion metaphors in classical poetry [Feng et al. 2024].)
- Speed $\Leftrightarrow$ Pace: Horizontal terrain movement correlates with recitation speed (faster speech - accelerated fluctuations).

Through this mapping, users "sculpt" unique poetic landscapes with their voices (Figure 3), receiving immediate visual feedback for real-time adjustment and training.

3 PRELIMINARY USER STUDY

To validate Poetry Space's efficacy and user experience, we conducted a pilot study with 8 university students (4 male, 4 female; avg. age 21) interested in classical Chinese poetry. Participants recited two poems with contrasting emotional tones. Semi-structured interviews and questionnaires revealed key feedback: (1) Enhanced motivation: Multiple participants noted that wanting to "paint" more beautiful mountainscapes motivated repeated practice (e.g., using slow, sorrowful delivery to form extended blue valleys for Thoughts on a Tranquil Night's homesickness). (2) Emotional connection: One participant stated: "I 'saw' the poem's emotion for the first time, feeling closer to the poet." (3) Intuitive embodied feedback: "As my voice rose and fell, so did the mountains. This real-time interaction heightened my focus on breath and pauses."

These results preliminarily validate Poetry Space's effectiveness in emotion visualization and real-time feedback for recitation training.

4 CONCLUSION

Poetry Space innovatively maps "sound-emotion-conception" in real-time, transforming invisible recitation emotions into perceptible artistic evolution. It constructs an immersive embodied feedback loop for recitation training, significantly enhancing learners' cultural aesthetic competence and cultural confidence. This approach offers museums, schools, and cultural institutions a viable, high-immersion solution for cultural experiences. Future work will integrate non-verbal cues (e.g., facial expressions, gestures) to accommodate diverse recitation training needs.

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